Modeling Volatility in Financial Markets

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1. Literature Review

1.1 Problem Origin

Financial volatility is the degree to which asset prices fluctuate over time and is central to risk management, derivative pricing, and portfolio allocation. The formal study of time-varying volatility began with Engle's (1982) ARCH model and was generalized by Bollerslev (1986) into the GARCH framework, which has dominated volatility research for four decades.

1.2 Problem Evolution

While GARCH-family models capture volatility clustering under a single regime, financial markets alternate between structurally distinct states that a single-regime specification cannot represent. Hamilton (1989) introduced the Markov-switching autoregressive model, and this insight was extended to volatility through MS-GARCH specifications (Haas et al., 2004; Klaassen, 2002). In parallel, machine learning methods—neural networks, recurrent architectures, and ensemble methods—were applied to forecast volatility directly, often outperforming classical models (Carr et al., 2020; Kim et al., 2018). More recently, ML techniques have been turned toward regime detection itself (Raja Lope Ahmad et al., 2026; Ejike, 2026).

1.3 Problem Impact

Underestimated tail risk contributed to catastrophic losses during the 2008 financial crisis, the 2020 COVID-19 crash, and the 2022 energy price shock. Improved volatility models enable more reliable Value-at-Risk estimates, better hedging strategies for energy-dependent firms, and more precise option pricing—serving both academic understanding and practical risk mitigation.

1.4 Previous Analytical Efforts

Approach 1: Markov-Switching GARCH Models. Haas et al. (2004) proposed a tractable MS-GARCH specification with independent conditional variance equations per regime, showing substantial improvements in density forecasts on foreign exchange data. Klaassen (2002) developed an alternative formulation adjusting expected variance at regime transitions to mitigate path-dependence. Ardia et al. (2019) provided the MSGARCH R package and confirmed regime-switching superiority in VaR forecasting on equity data. In commodity markets, Alizadeh et al. (2008) applied Markov-switching stochastic volatility to crude oil, Charles and Darné (2017) demonstrated MS-GARCH outperformance during crisis periods, and Hardy (2001) showed similar improvements for the S&P 500. **Merits:** grounded in statistical theory, interpretable regime probabilities, strong VaR improvements. **Shortcomings:** computational complexity grows with regime count, the number of states must be pre-specified, and these models do not easily incorporate high-dimensional features.

Approach 2: ML-Based Volatility Prediction. Carr et al. (2020) used Ridge Regression, Random Forest, and feedforward neural networks to predict volatility and construct a volatility index. Doering et al. (2017) applied CNNs to high-frequency data with strong performance but limited interpretability. Recurrent architectures have been explored more extensively: Tino et al. (2001) combined Markov models with an Elman RNN for volatility-based options trading, Xiong et al. (2016) integrated price data with Google Domestic Trends in an LSTM framework, and Kim et al. (2018) developed LSTM models integrated with GARCH. **Merits:** capture complex nonlinear patterns, incorporate diverse features, and outperform traditional models in point accuracy. **Shortcomings:** black-box problem limits theoretical insight, prone to overfitting, and lack the probabilistic regime interpretation of MS-GARCH.

Approach 3: ML-Based Regime Detection. Raja Lope Ahmad et al. (2026) combined PCA and clustering with attention-based RNNs to detect market instability shifts without pre-specifying the number of states. Ejike (2026) compared LDA, SVMs, and feedforward neural networks for regime classification, finding nonlinear classifiers offered improved accuracy. **Merits:** data-driven regime discovery from high-dimensional features without parametric assumptions. **Shortcomings:** regime labels are often determined ex post, require large training samples, and lack the transition-probability framework of Markov-switching models.

2. The Plan

2.1 Rationale

We chose this project because volatility modeling aligns with the core themes of this course, and SPY and crude oil offer complementary profiles: SPY exhibits mean-reverting volatility with leverage effects, while crude oil is driven by supply shocks producing abrupt regime shifts (Alizadeh et al., 2008). Our methods span three tiers—traditional GARCH, Markov-switching GARCH, and LSTM—to evaluate whether added complexity yields forecasting gains. Key challenges include regime count selection, lookahead bias prevention, and obtaining reliable realized volatility benchmarks. The current US-Iran conflict has also led to a major disruption in the oil markets - and modeling these fluctuations would be fruitful for and further prediction during such shocks in the system.

2.2 Dataset Selection

We plan to use two daily price series retrieved from Yahoo Finance: the SPDR S&P 500 ETF Trust (SPY) and West Texas Intermediate crude oil futures (CL=F), spanning Jan 2001 through Dec 2025. This window encompasses multiple known volatility regimes—the 2008 financial crisis, the 2014–2016 oil price collapse, the COVID-19 crash, and the 2022 energy shock—ensuring sufficient regime variation for estimation and evaluation. Log returns are computed as $r_t=\ln(P_t/P_{t-1})$ and serve as the primary modeling input. Ljung–Box tests on squared residuals confirm the presence of conditional heteroskedasticity (See EDA).

2.3 Approach 1 – Traditional GARCH Models

As a baseline, we will fit GARCH(1,1) and GJR-GARCH(1,1) models to both return series using maximum likelihood estimation, implemented via the *arch* Python package. The standard GARCH~~(1,1)~~ captures symmetric volatility persistence (Bollerslev, 1986), while the GJR-GARCH~~(1,1)~~ incorporates a leverage term to model the asymmetric response of volatility to positive versus negative shocks (Glosten et al., 1993). Model adequacy will be assessed via Ljung–Box tests on standardized residuals, ARCH-LM tests for remaining heteroskedasticity, and QQ plots to evaluate distributional fit.

Out-of-sample evaluation will be a rolling-window one-step-ahead forecasting design. Forecast accuracy will be measured by RMSE and MAE of conditional variance forecasts against squared returns as a realized volatility proxy.

2.4 Approach 2 – Markov-Switching GARCH Models

We'll extend the baseline by estimating two-state and three-state MS-GARCH(1,1) models with both normal and Student-*t* innovations. Following Haas et al. (2004), each regime maintains an independent GARCH variance equation, allowing the model to capture structurally different volatility dynamics across calm and turbulent states. Estimation will be performed via the MSGARCH R package (Ardia et al., 2019), called from Python through *rpy2*. Model selection among competing specifications (number of regimes, innovation distribution) will be guided by AIC, BIC, and log-likelihood comparisons.

Out-of-sample evaluation will be similar to the rolling-window design from Approach 1. In addition, we'll conduct Value-at-Risk backtesting at the 1% and 5% levels using the Kupiec unconditional coverage test and the Christoffersen conditional coverage test, as VaR accuracy

is a key advantage of regime-switching models over single-regime GARCH (Ardia et al., 2019).

2.5 Approach 3 – LSTM-Based Volatility Prediction

In the third approach we'll use Long Short-Term Memory networks to forecast volatility directly from sequential features. Following Kim et al. (2018), who demonstrated the value of integrating LSTM with GARCH-type inputs, we construct feature vectors consisting of lagged log returns, lagged squared returns, rolling realized volatility estimates (5-, 10-, and 21-day windows), and the VIX index (for SPY) and OVX index (for crude oil) as implied volatility proxies.

A strict temporal split will be enforced: training (2001–2018), validation (2019–2021), and test (2022–2025) periods. For evaluation we'll use the same RMSE, MAE, and VaR backtesting metrics as the previous approaches. This will allow us to assess whether the added complexity and reduced interpretability of deep learning yields meaningful forecasting gains over the statistically grounded GARCH and MS-GARCH alternatives (Xiong et al., 2016).

2.6 Challenges anticipated

A key challenge is the selection of the correct number of states/regimes for the MS-GARCH model, and modeling sudden shocks in the system and making sure that the residual is white noise. Learning from inferences from previous shocks and incorporating it into our model.

3. Exploratory Data Analysis

3.1 Stationarity

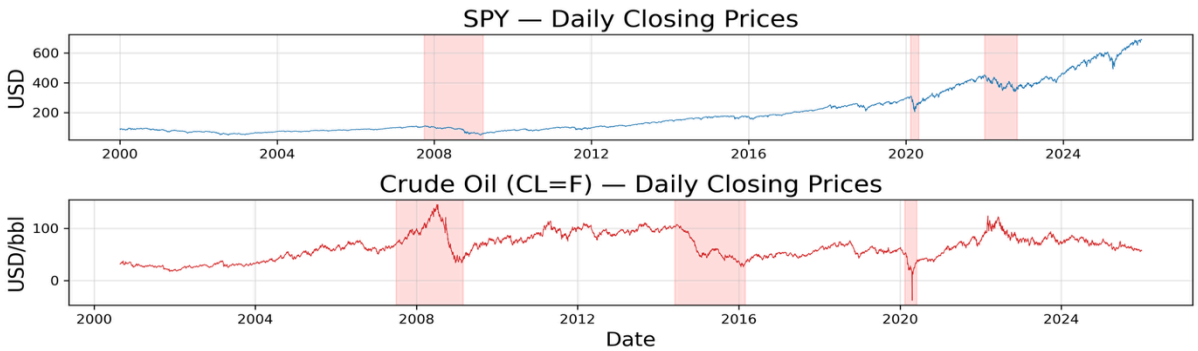

Fig. 1: SPY and OIL time series plot

For both assets, the ACF decays extremely slowly with all lags through 40 near 1.0 and well above the confidence band, and the PACF shows a single dominant spike at lag 1 with all subsequent lags near zero. This shows both series are non-stationary.

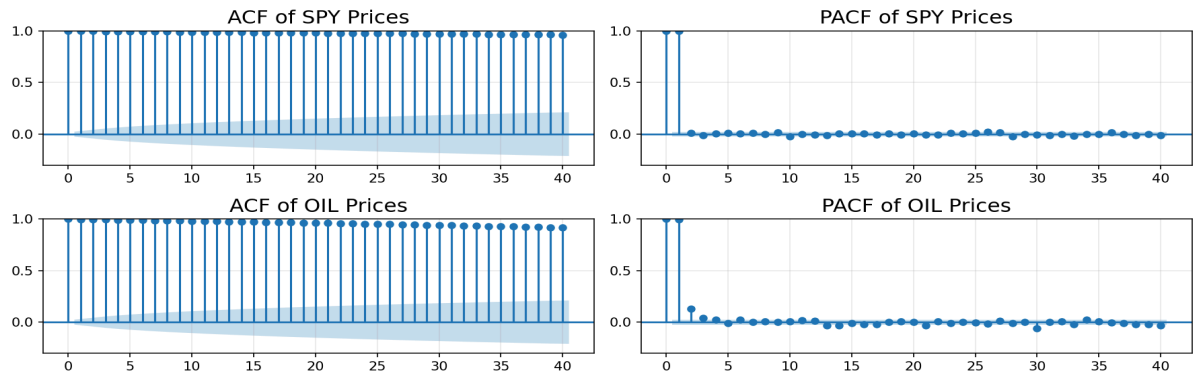

Fig. 2: ACF, PACF of SPY and OIL prices

3.2 Best ARIMA Fit and Squared Residual Analysis for heteroskedasticity

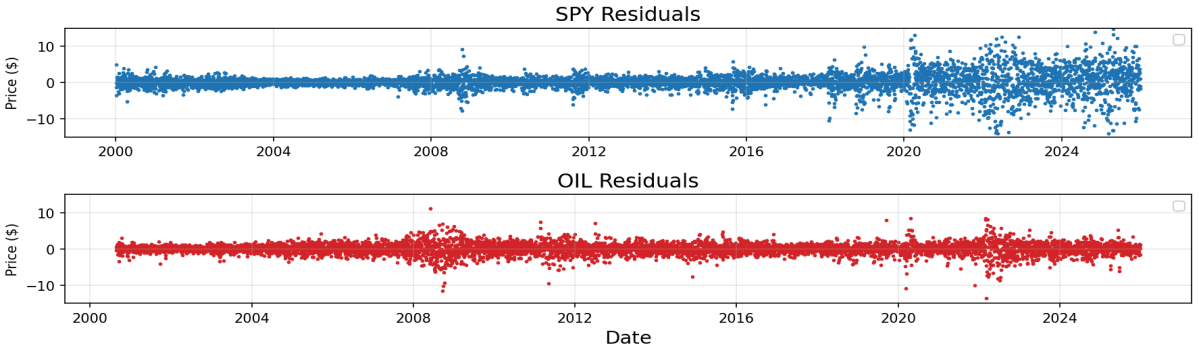

Fig. 3: SPY and OIL residuals after best ARIMA fits – ARIMA(7,1,7) for SPY and ARIMA(6,1,6) for OIL

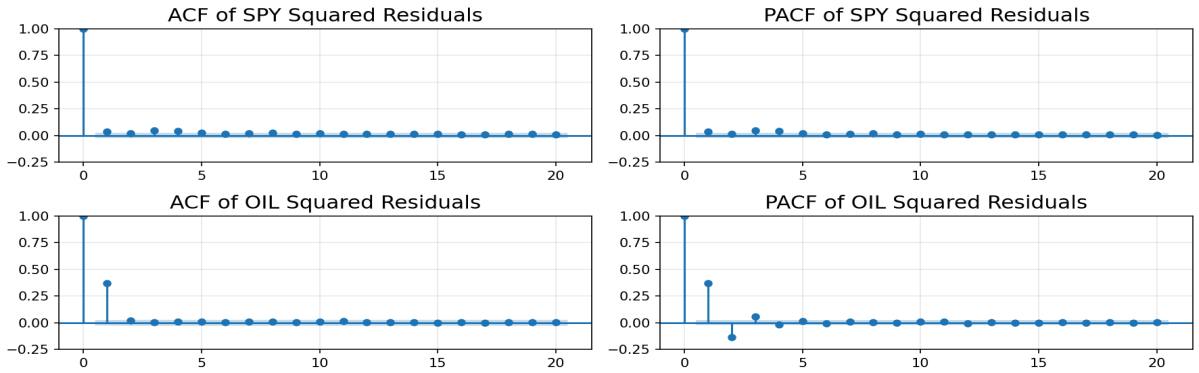

Fig 4: ACF, PACF of SPY & OIL residuals after best ARIMA fits

Asset	Lag	LB statistic	p-value	Significant
SPY	10	51.459	1.44×10^{-7}	Yes
SPY	20	60.948	5.08×10^{-6}	Yes
SPY	30	72.075	2.55×10^{-5}	Yes
OIL	10	876.824	6.19×10^{-182}	Yes
OIL	20	878.817	2.53×10^{-173}	Yes
OIL	30	887.966	2.08×10^{-167}	Yes

Table 1: Ljung–Box tests for heteroskedasticity on squared residuals

Both assets show overwhelmingly significant ARCH effects at all lag lengths, confirming conditional heteroskedasticity in the squared residuals. OIL's test statistics are about 17× larger than SPY's, reflecting much more persistent volatility clustering in crude oil. This analysis strengthens the need for stronger variance/volatility modelling.

4. Appendix: Team Contributions

Contributions

[Rajath] – Report writing, Report assembly, Review member work
[Harsh] – Report writing, Report assembly, Review member work
[Yibo] – Report writing, Report assembly, Review member work
[Mani] – Report writing, Report assembly, Review member work
[Yen-Shuo] – Report writing, Report assembly, Review member work

Approach Attribution

[Rajath] – Traditional ARCH/GARCH models, exploratory data analysis, data preprocessing
[Harsh] – Regime-switching MS-GARCH models, regime probability analysis
[Yibo] – Literature review on machine learning approaches, comparative evaluation
[Mani] – Literature review, exploratory data analysis
[Yen-Shuo] – Literature review on VIX, exploratory data analysis

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